

Intelligent 2/1 problem

- ✓ Problem Solving through intelligence
- ✓ Two-One Problem
 - ✓ Problem formulation
- ✓ Decision Tree
- ✓ Problem Solving search space
- ✓ Graphs
- ✓ Trees

Two – One Problem

Start

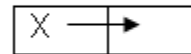
1 1 ? 2 2

Goal

2 2 ? 1 1

Legal Moves

Slide



Hop



Rules:

- 1s' move right
- 2s' move left
- Only one move at a time
- No backing up

Legal Moves:

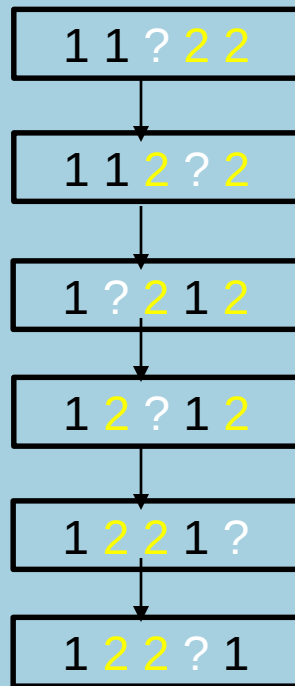
- Slide
- Hop



Two – One Problem

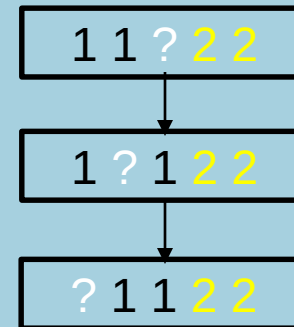
Trials to solve the problem

Trial One



Stuck!!!

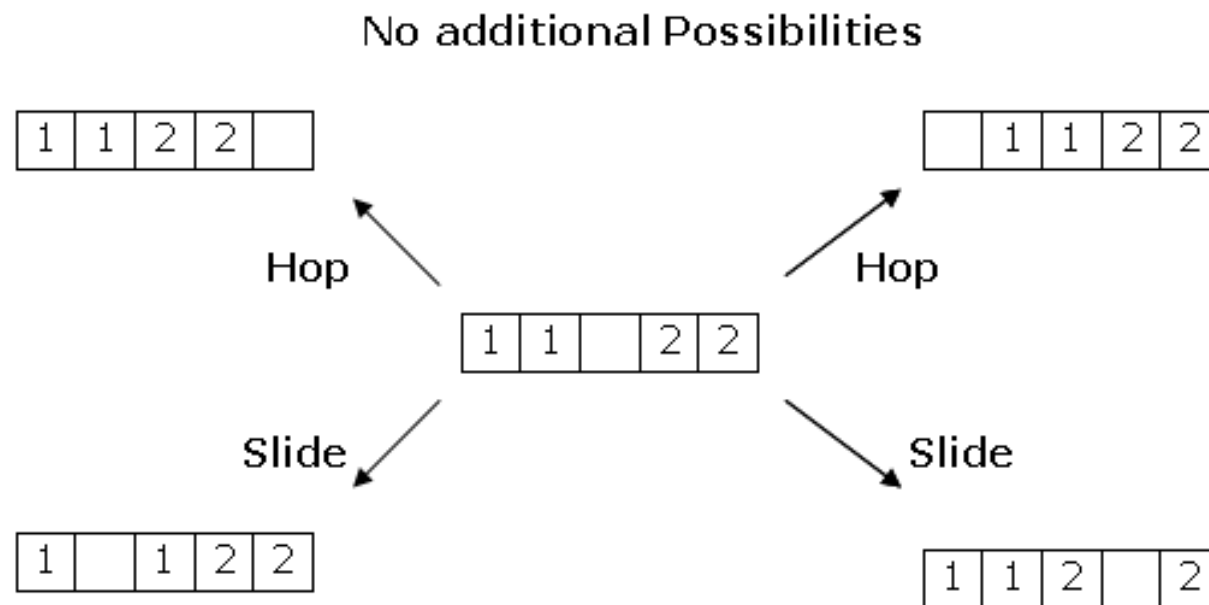
Trial Two



Stuck!!!

Two – One Problem

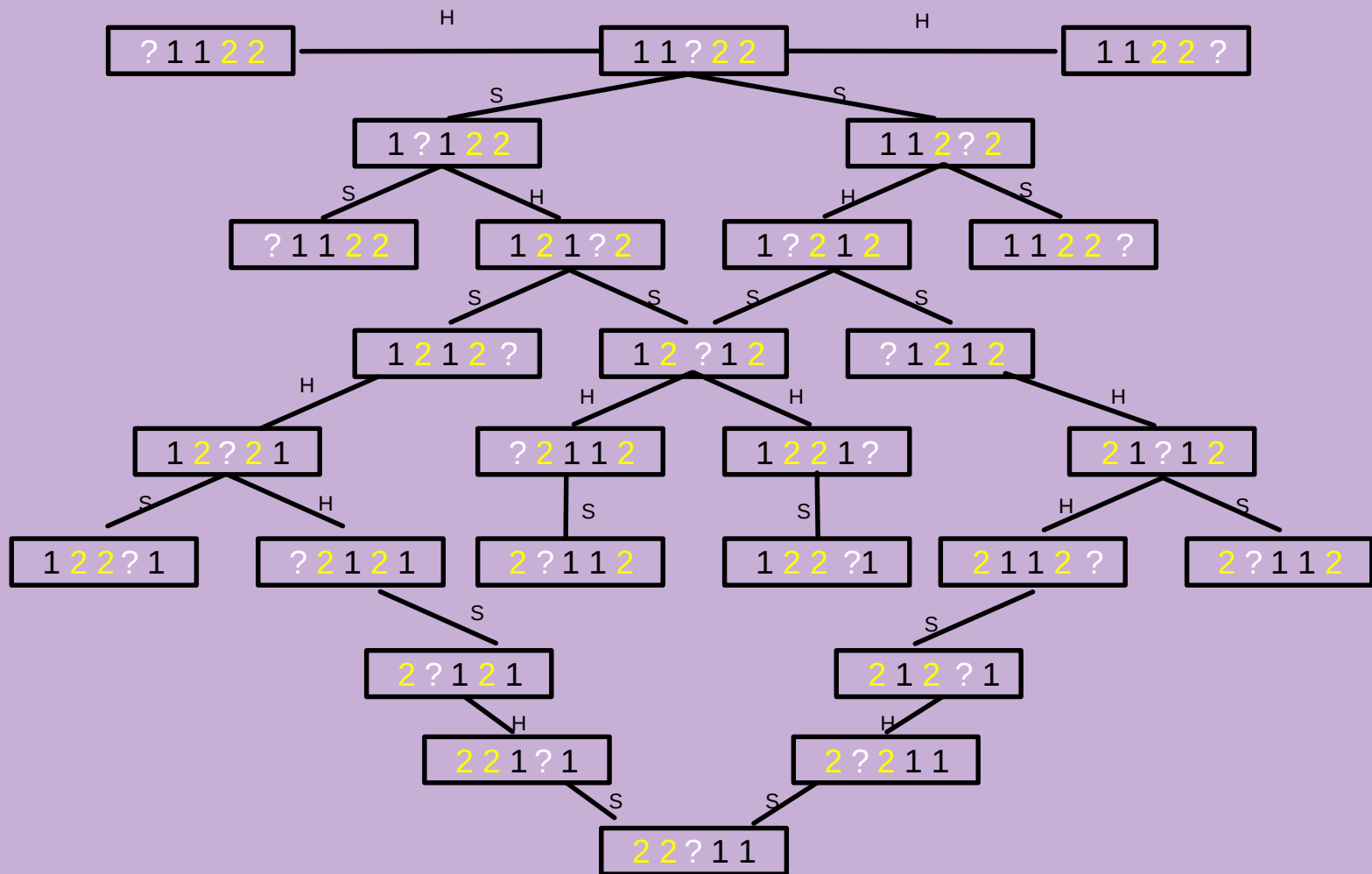
Five States



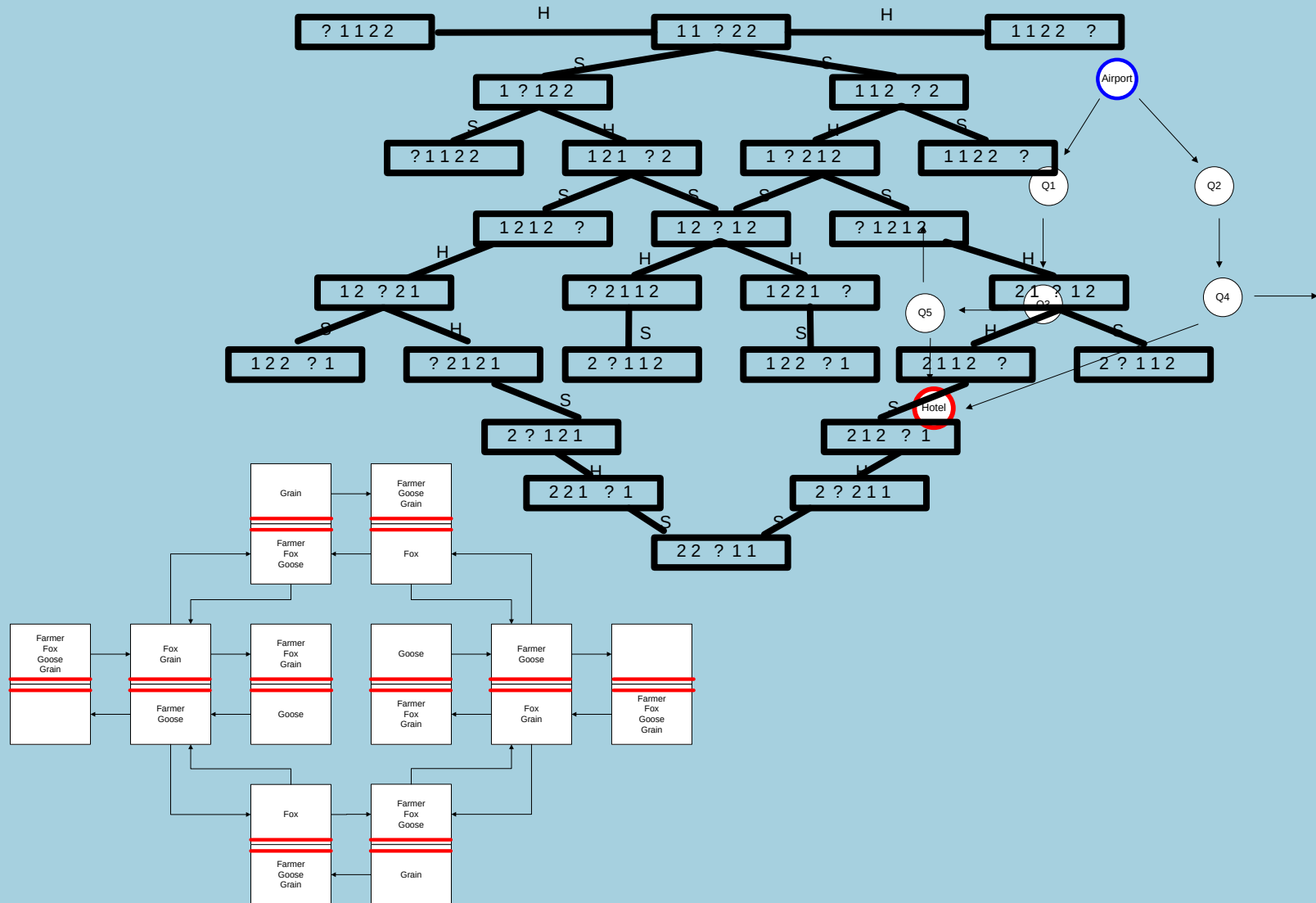
| Both hopping and sliding can still be applied

Two – One Problem

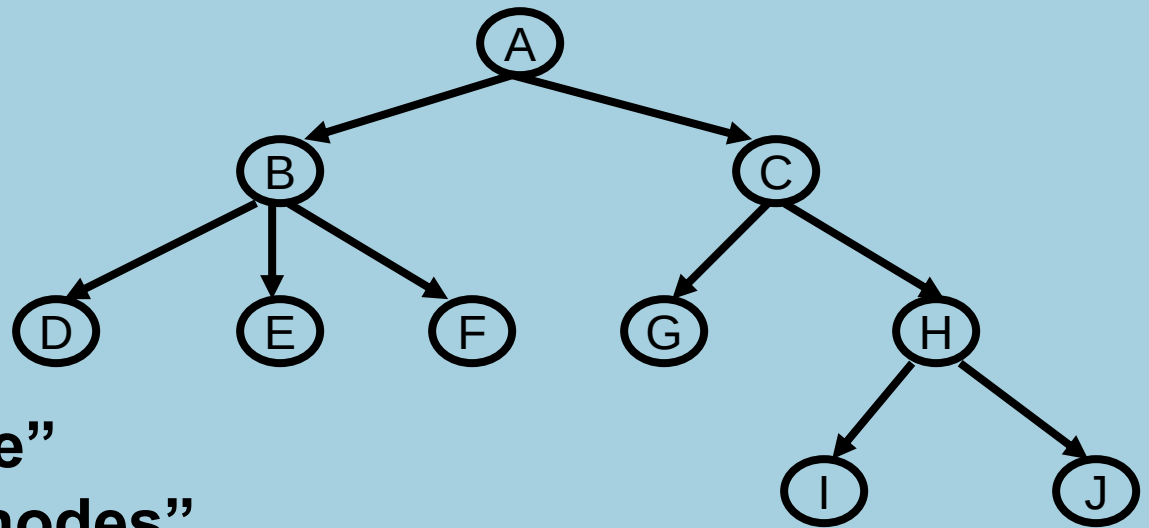
Solution Space



Solution to Problem Solving : Searching



Tree and Graph Terminology

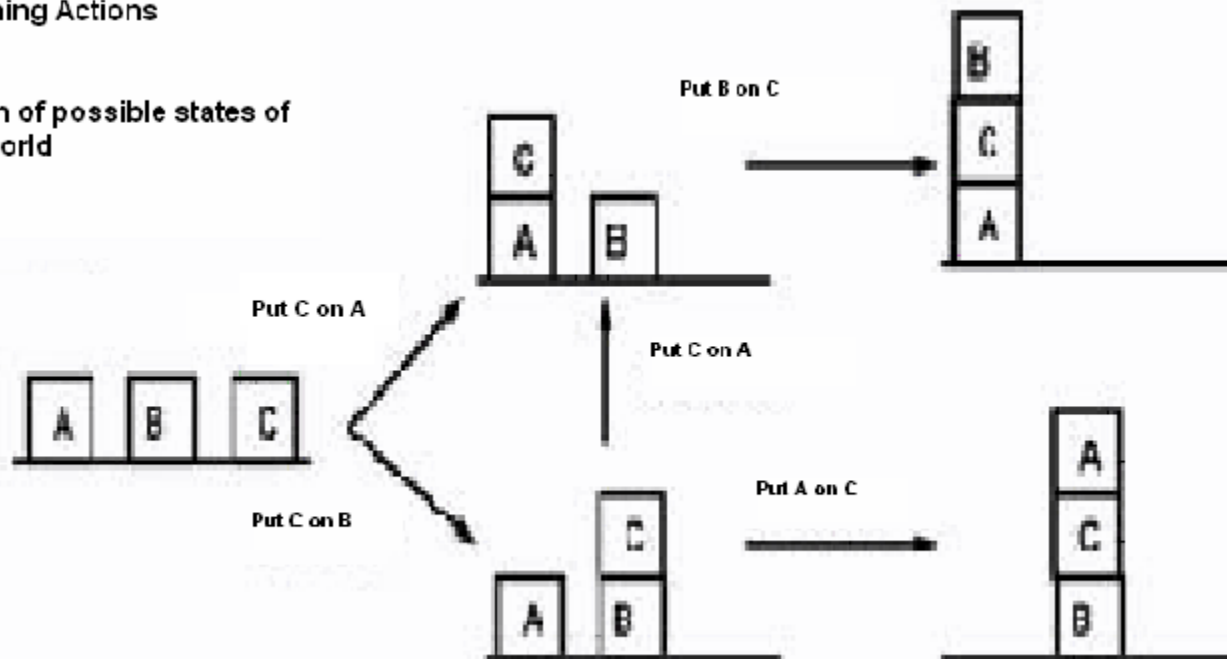


- “A” is the “root node”
- “A, B, C J” are “nodes”
- “B” is a “child” of “A”
- “A” is ancestor of “D”
- “D” is a descendant of “A”
- “D, E, F, G, I, J” are “leaf nodes”
- Arrows represent “edges” or “links”

Examples of Graphs

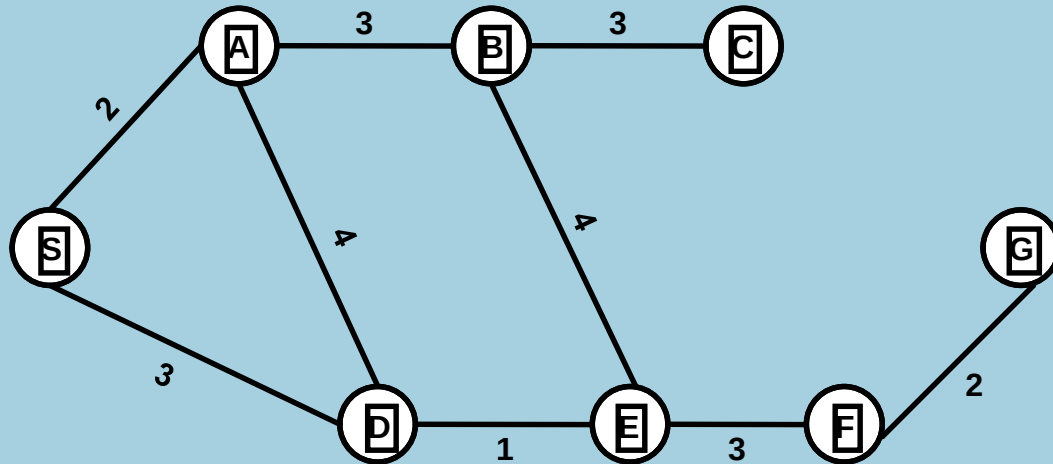
Planning Actions

Graph of possible states of the world



Problem Formulation using Graphs

- ▣ The search methods we'll be dealing with are defined on trees and graphs



Tree Search

- Graph search is really tree search

